

# CareerAI Copilot

Personalized AI Career & Job Readiness Assistant

|              |                                                         |
|--------------|---------------------------------------------------------|
| Hackathon    | SkillUp Hackathon in collaboration with IBM SkillsBuild |
| Track        | Student AI Track                                        |
| Team         | CareerForge AI                                          |
| College      | Pragati Engineering College                             |
| Project Lead | Aganti Srinidhi                                         |

## Project Vision

Help students turn their existing skills, projects, and career interests into a clear and actionable path toward career readiness.

Resume / GitHub → Skill Analysis → Career Recommendation → Skill Gap → Learning Roadmap → Mock Interview → Readiness Score

# 1. Project Overview

**CareerAI Copilot** is an AI-powered career guidance concept designed for college students. It brings profile analysis, career matching, skill-gap identification, learning guidance, interview practice, and job-readiness feedback into one workflow.

The project addresses a common student challenge: having multiple skills and projects without a clear understanding of which career path fits best or which skills should be developed next.

## Objectives

- Identify skills from a student's resume and project information.
- Recommend a suitable career path based on the student's profile.
- Highlight missing skills for the selected career.
- Convert skill gaps into a structured learning roadmap.
- Provide career-focused mock interview practice and feedback.
- Present an understandable job-readiness score.

## 2. Problem Statement

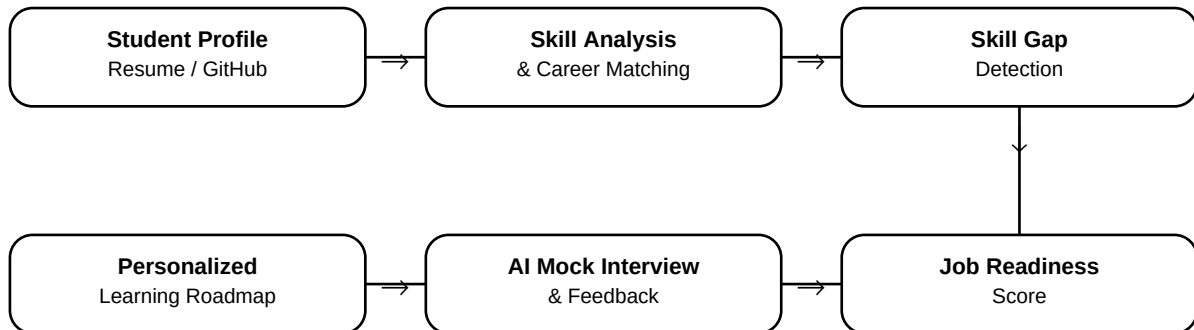
Many college students struggle to identify the right career path and understand the skills required to become job-ready. Students may have resumes, GitHub projects, and access to learning resources, but often lack personalized guidance connecting their current profile to a specific career goal. This can lead to unfocused learning, difficulty identifying skill gaps, and insufficient interview preparation.

## 3. Proposed Solution

CareerAI Copilot provides a single career-readiness workflow. A student supplies resume/skills information and optional GitHub project details, selects a career goal or allows the system to recommend one, and receives a profile-based analysis. The current prototype presents detected skills, career matching, missing skills, a personalized roadmap, mock interview questions, interview feedback, and a readiness score.

## 4. System Workflow & Architecture

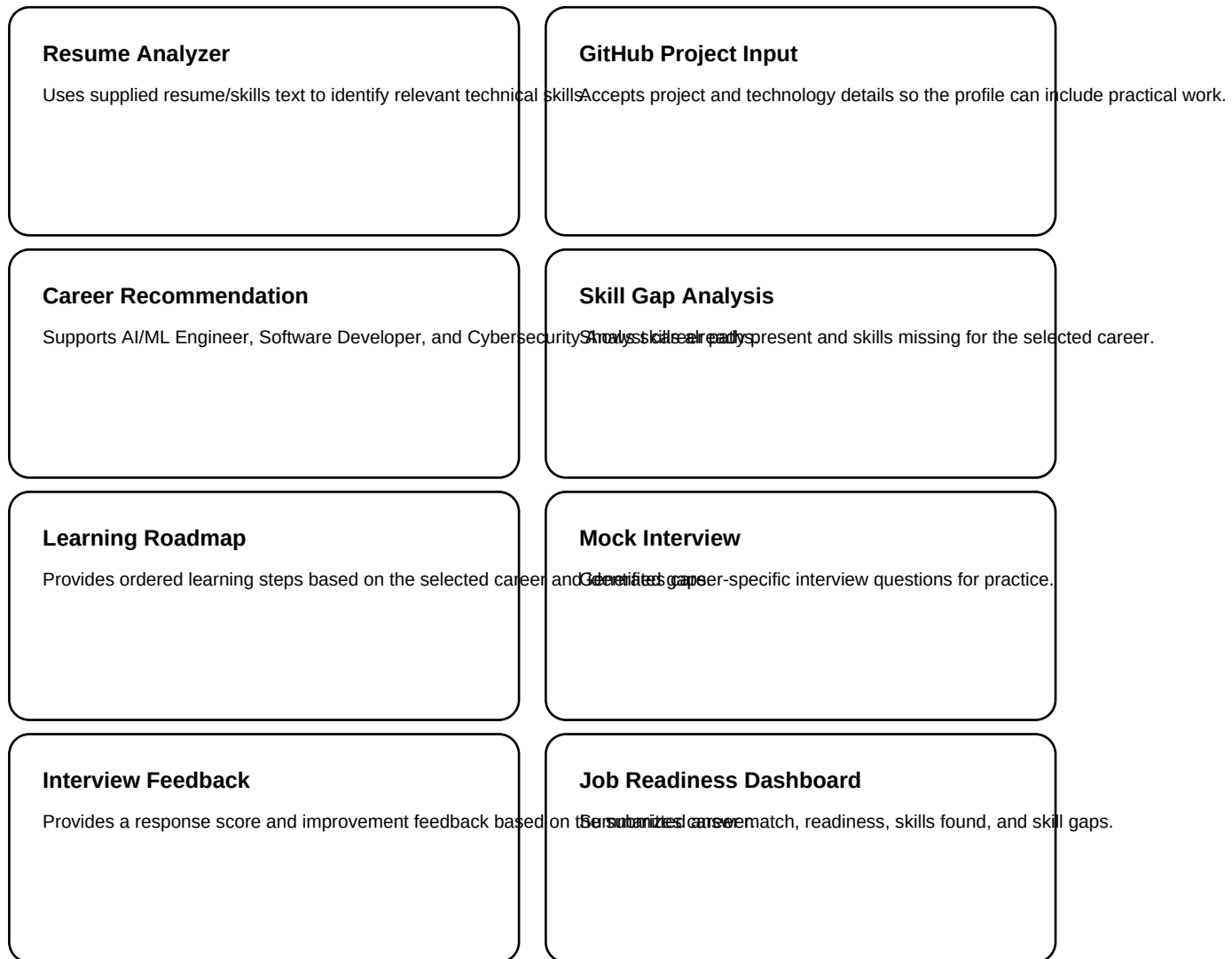
The project follows a sequential career-readiness workflow that turns student information into actionable outputs.



### Workflow Explanation

|   |                 |                                                                                       |
|---|-----------------|---------------------------------------------------------------------------------------|
| 1 | Student Input   | Resume/skills, GitHub project details, and optional career goal.                      |
| 2 | Skill Analysis  | Extract relevant technical skills from the supplied profile information.              |
| 3 | Career Matching | Compare student skills with the skill set associated with supported career paths.     |
| 4 | Skill Gap       | Show matched skills and missing skills for the selected career.                       |
| 5 | Roadmap         | Generate a sequence of learning and project-building steps.                           |
| 6 | Interview       | Generate career-specific questions and evaluate submitted answers.                    |
| 7 | Readiness       | Present a simple readiness indicator based on the current prototype's matching score. |

## 5. Key Features



## 6. Technology Stack

| Layer               | Technology / Approach                                              |
|---------------------|--------------------------------------------------------------------|
| Frontend            | HTML, CSS, JavaScript                                              |
| Backend             | Python with Flask                                                  |
| Application Logic   | Career matching, skill extraction, roadmap and interview workflows |
| Development Support | IBM Bob                                                            |
| Version Control     | GitHub                                                             |

## 7. IBM Bob Usage

IBM Bob was used as an AI-powered development assistant during the project lifecycle. Its role in the project was to support planning, application development, AI workflow design, debugging, testing, interface refinement, and documentation.

| Development Activity    | How IBM Bob Supported the Project                                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Planning & Architecture | Helped break the idea into modules and structure the end-to-end workflow.                                   |
| Code Development        | Assisted with generating, explaining, refining, and organizing application code.                            |
| AI Workflow Design      | Supported design of prompts and workflows for career matching, skill gaps, roadmaps, and interviews.        |
| Debugging               | Helped identify programming issues and suggest fixes during development.                                    |
| Testing                 | Suggested test scenarios and possible edge cases for application workflows.                                 |
| UI/UX                   | Assisted with dashboard structure and presentation of career-readiness information.                         |
| Documentation           | Assisted with project explanations, workflow descriptions, README content, and IBM Bob usage documentation. |

### IBM Bob Development Workflow

Planning → Architecture → Code Generation → AI Workflow Development → Debugging → Testing → UI Refinement  
→ Documentation

**Responsible development:** AI-generated code and suggestions were reviewed and adapted by the development team before being incorporated into the project.

## 8. Demonstration Flow

The project demo is designed to show the complete student journey from profile input to career-readiness guidance.

| Stage | Demo Action                                       | Expected Output                                            |
|-------|---------------------------------------------------|------------------------------------------------------------|
| 1     | Enter resume/skills                               | Profile information is supplied to the system.             |
| 2     | Enter GitHub/project details                      | Additional project and technology information is included. |
| 3     | Select AI/ML Engineer or let the system recommend | Career path is selected.                                   |
| 4     | Click Analyze My Career                           | Career match, readiness, skills, and gaps appear.          |
| 5     | Review roadmap                                    | Ordered learning steps are displayed.                      |
| 6     | Start mock interview                              | Career-specific questions are generated.                   |
| 7     | Submit an answer                                  | Score and improvement feedback are shown.                  |

## 9. Expected Impact

CareerAI Copilot is intended to help students make more informed career decisions by connecting their current skills with possible career paths and concrete next steps. The project combines guidance and preparation into one student-friendly workflow.

## 10. Future Enhancements

- Direct GitHub profile integration for automated repository analysis.
- Broader career-domain coverage and richer career matching.
- Integration of verified learning resources, including relevant SkillsBuild content.
- More advanced resume scoring and job recommendations.
- Multilingual and voice-based career assistance.
- Student progress tracking over time.

## 11. Conclusion

CareerAI Copilot demonstrates a practical AI-assisted approach to student career readiness. By combining profile analysis, career matching, skill-gap identification, learning guidance, and interview practice, the project provides a structured path from a student's current profile toward a target career.

### Submission Assets

GitHub Repository: To be added after the public repository is created.

Demo Video: CareerAI Copilot hackathon demonstration.

IBM Bob Documentation: IBM\_BOB\_USAGE.md in the GitHub repository.